THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In an experiment 5.6 g of CaO reacts with 4.4 g of CO2 to form CaCO3. The mass of CaCO3 formed is: 1

(a) 1.2 g **(b)** 9.6 g **(c)** 11 g **(d)** 10 g

2. The formula of aluminium sulphate is: 1

(a) $Al_3(SO_4)_2$ **(b)** $Al_2(SO_4)_3$ **(c)** $Al(SO_4)_3$ **(d)** $AlSO_4$

3. Which of these pairs forms an ionic compound? 1

(a) C and O **(b)** Mg and Cl **(c)** N and H **(d)** C and H

4. Ammonia contains 82.35 per cent nitrogen. In 34 g of ammonia the mass of nitrogen is: 1

(a) 6 g **(b)** 28 g **(c)** 14 g **(d)** 17 g

5. The formula of ammonium phosphate is: 1

(a) NH_4PO_4 **(b)** $(NH_4)_3PO_4$ **(c)** $NH_4(PO_4)_3$ **(d)** $(NH_4)_2PO_4$

6. Which of the following substances will conduct electricity in the solid state? 1

(a) sodium chloride **(b)** copper **(c)** sulphur **(d)** sugar

7. A compound has 40 per cent sulphur and 60 per cent oxygen by mass. 20 g of sulphur combines with how much oxygen? 1

(a) 12 g **(b)** 30 g **(c)** 40 g **(d)** 60 g

8. The formula of calcium hydroxide is: 1

(a) Ca_2OH **(b)** $Ca(OH)_2$ **(c)** CaO_2H **(d)** $CaOH$

9. Which compound has a double bond between its atoms? 1

(a) Cl_2 **(b)** O_2 **(c)** N_2 **(d)** H_2

10. Carbon monoxide has C : O = 3 : 4. The mass of oxygen that combines with 9 g of carbon is: 1

(a) 16 g **(b)** 12 g **(c)** 6.75 g **(d)** 9 g

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In an experiment 2.0 g of hydrogen reacts with 16 g of oxygen to form water. Find the mass of water and verify the law of conservation of mass. 2

12. Why is the formula of water not HO? Explain with valency. 2

13. Calculate the percentage of carbon in CO_2. 2

14. Calculate the mass of hydrogen and oxygen in 36 g of water. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A compound of carbon and oxygen has C : O = 3 : 8 by mass. Find the amounts of C and O in 55 g of the compound. 3

16. State three differences between sodium chloride and carbon tetrachloride. 3

17. An element X forms an oxide X_2O_3 with 70 per cent X by mass. Find the atomic mass of X (O = 16). 3

OR

Magnesium (4.8 g) combines with nitrogen to form 6.6 g magnesium nitride. Find the mass of nitrogen used and the ratio Mg : N. 3

18. Explain why an ionic solid does not conduct but its melt does. 3

OR

Calculate the mass of calcium oxide produced from 200 g of calcium carbonate if 88 g of CO2 is given out. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Verify the law of multiple proportions with carbon and oxygen: CO has 12 g C with 16 g O, CO2 has 12 g C with 32 g O. Explain the ratio and the role of Dalton's theory. 5

OR

A student heats 6.0 g of copper in air. The mass of the product copper oxide is 7.5 g. Find the mass of oxygen, the ratio of Cu : O and compare with the formula CuO (Cu = 63.5, O = 16). 5

20. Explain how the electronic structure explains the formation of ionic compounds NaCl, MgO and CaF2 with the charges and the formulae. 5

OR

Explain how scientists determined the relative atomic masses from the laws of chemical combination and give the reference standard used. 5

END OF QUESTION PAPER Total: 40 Marks

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